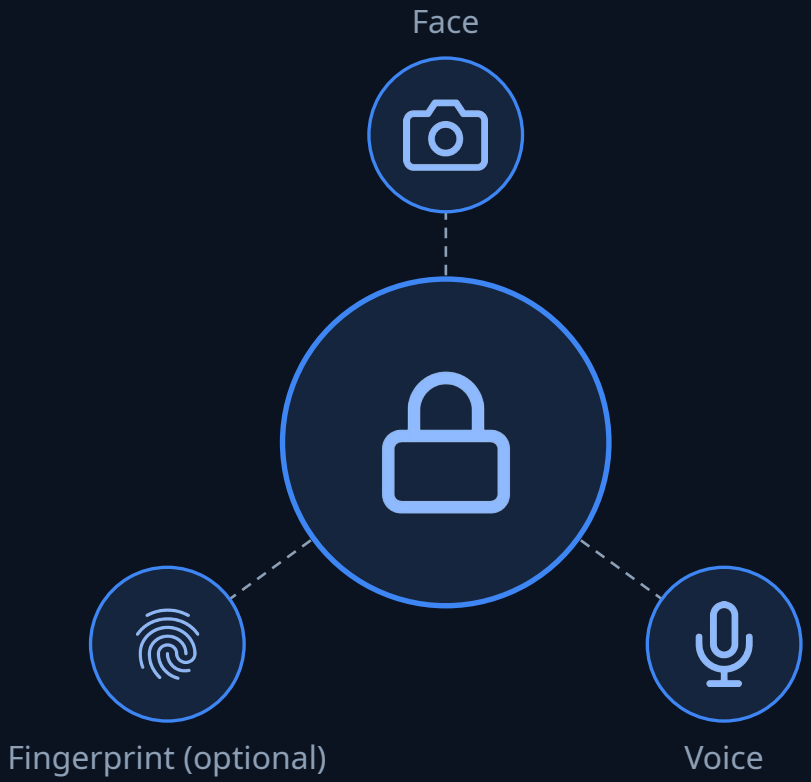
FINAL YEAR PROJECT REVIEW

TriLock

Risk-adaptive multimodal biometric authentication for digital banking



Face | Voice | Fingerprint

Academic prototype. All transactions are simulated.

Passwords alone do not prove who you are

Credentials can be taken

Passwords and one-time codes can be guessed, phished or shared. Whoever has them is "you".

One biometric is not enough

A single trait can be faked with a photo or a recording, or fail in bad light and noise.

Every payment gets the same check

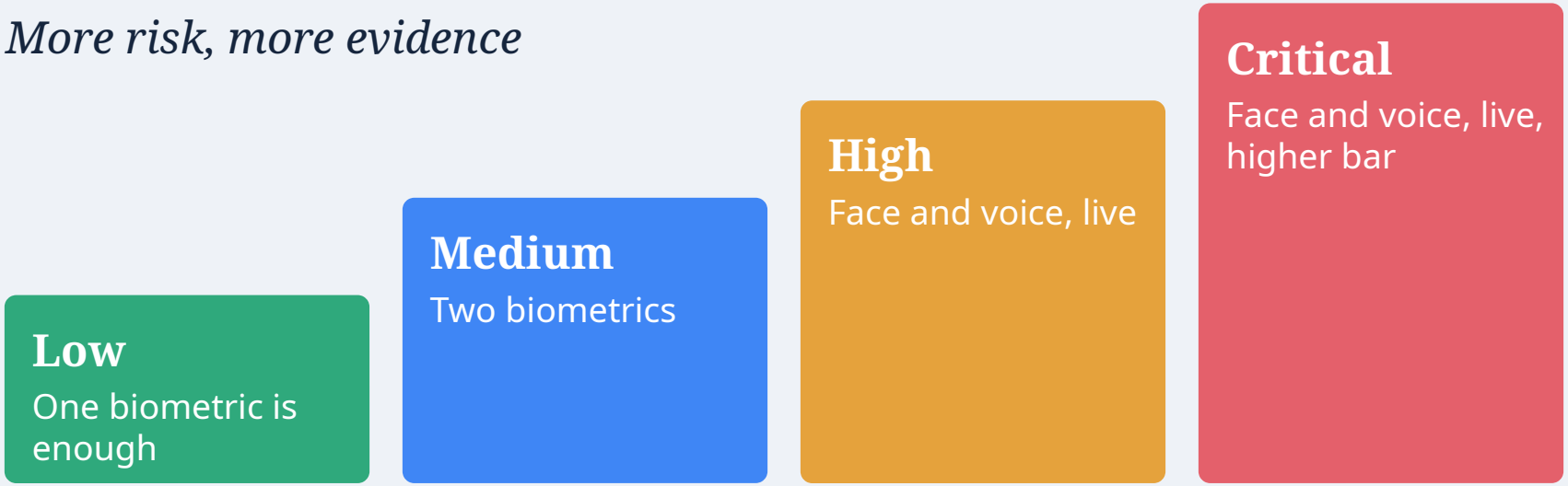
A coffee and a large transfer are treated alike, so the check is either too weak or too annoying.

The question

How do we verify a person strongly, without making every small payment painful?

The idea: match the check to the risk

More risk, more evidence

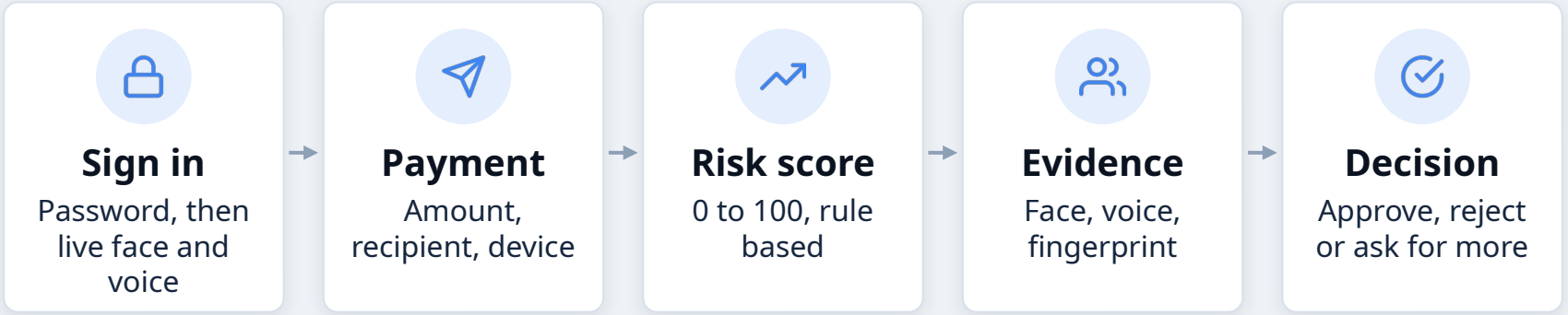


Small payments stay smooth

Large payments get more proof

Missing evidence is never a match

TriLock in one view



Multimodal

Three traits that fail in different ways, so they cover each other.

Risk-adaptive

The bar rises with the risk of the payment.

Liveness-aware

High-risk steps need a live head turn and spoken words.

System architecture

Browser
React + Vite

- Camera and microphone
- Dashboard and analytics
- Makes no decisions

API
FastAPI

- Routes, validation
- Sessions, access control
- One error format

Services
Python

- Risk engine
- Fusion and policy
- Sign-in and liveness

Biometrics
3 modules

- Face: YuNet, SFace
- Voice: ECAPA-TDNN
- Finger: SIFT, RANSAC

SQLite
users, payments, attempts, audit log

Templates
encrypted numbers, never photos

Evaluation
offline scripts, reproducible

Why face, voice and fingerprint



Face

Required

Every laptop and phone has a camera. It is quick, natural and needs no extra hardware.



Voice

Required

Works remotely with only a microphone, and carries different information than the face.



Fingerprint

Optional

A strong trait, but a browser cannot read a sensor. We work on images, so it is a future enhancement.

Different traits fail in different ways, so combining them covers each other's weak spots.

How a face or voice is recognised

1 Sample	2 Quality gate	3 Embedding	4 Compare	5 Confidence
A camera frame or an audio clip	Face found, sharp, lit. Speech present, not clipped	128 numbers for a face, 192 for a voice	Cosine similarity to the enrolled reference	Calibrated to a 0 to 1 match probability

What is stored

Only the numbers, encrypted. Photos and recordings are processed in memory and never saved.

Why calibrate

Face and voice scores mean different things, so each model is mapped to a match probability before we combine them.

Scoring how risky a payment is

What adds risk (points)

Amount	**0 to 40**
New recipient	**+20**
International	**+12**
New device	**+10**
Unusual amount for this user	**+12**
Several payments in minutes	**+10 / +20**
Unusual hour for this user	**+8**
New account, large amount	**+10**
Biometrics changed recently	**+12**

Four risk levels

Low	Medium	High	Critical
under 25	25 to 49	50 to 74	75 and up

Rule based and explainable: every point is shown to the user. It is not a trained fraud model.

The time factors learn your usual hours from your approved payments.

Combining the evidence into a decision

Evidence
per modality

→

Weigh by quality
poor input counts less

→

Fused score
plus coverage

→

Risk sets the bar
thresholds per level

Risk	Approve at	Evidence needed
Low	0.60	any 1 biometric
Medium	0.70	any 2 biometrics
High	0.80	face and voice
Critical	0.88	face and voice

Approved evidence is strong enough

Rejected evidence clearly does not match

Ask for more not enough evidence yet

A missing or failed input is never counted as a match. Fingerprint only adds evidence when present.

Security and privacy built in

Strong sign-in

Password, then live face and voice.

Liveness checks

Head turn and three random spoken words.

Smart lockouts

An attacker's failures cannot lock you out.

Protected enrollment

Changing biometrics needs the password and a matching sample.

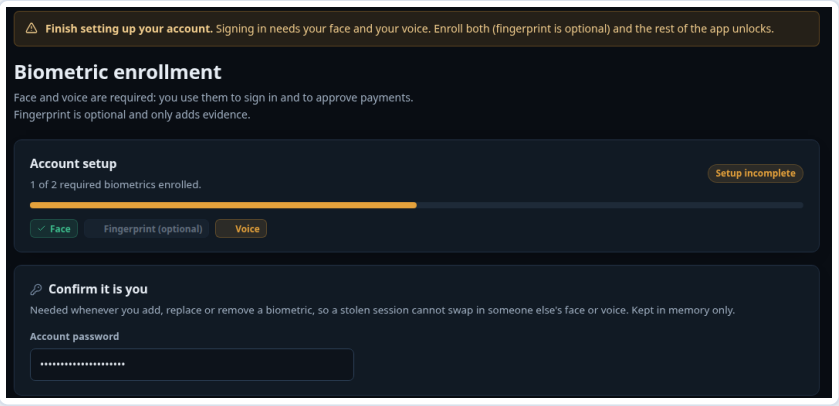
Careful recovery

Waiting period, admin approval, then probation.

Privacy by default

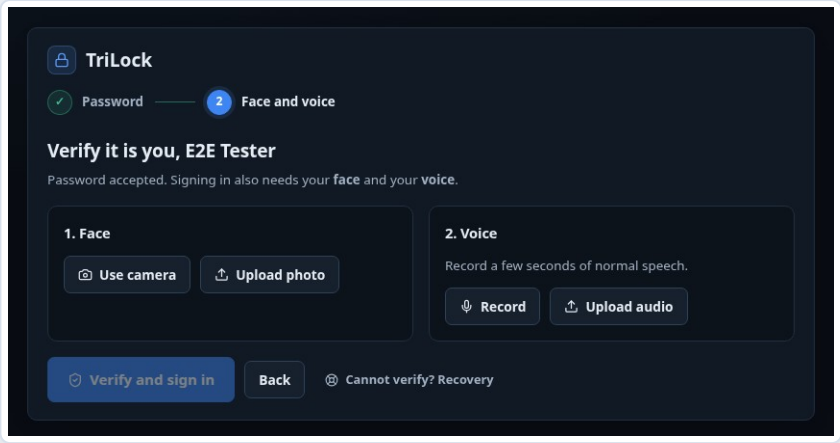
No raw photos or recordings stored. Templates encrypted.

Setting up and signing in



Enrollment

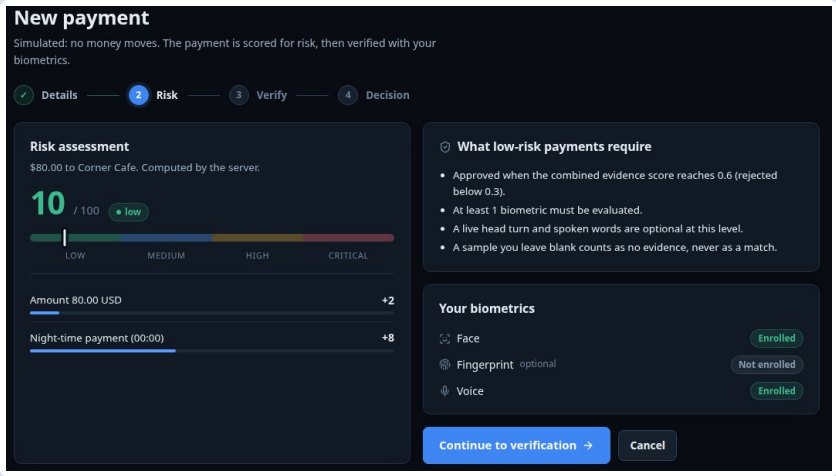
Face and voice are required, fingerprint is optional. A reference only counts once the server has built it.

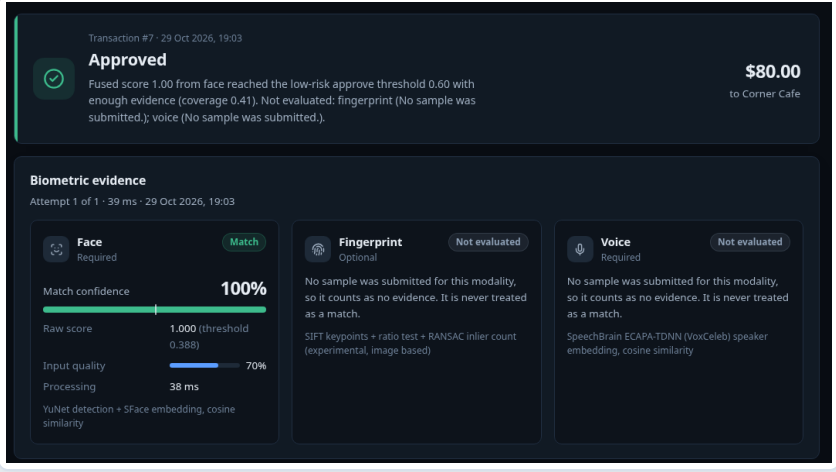


Sign-in

Password first. Then a live face and voice check, with a recovery option if it cannot be passed.

A small payment, start to finish





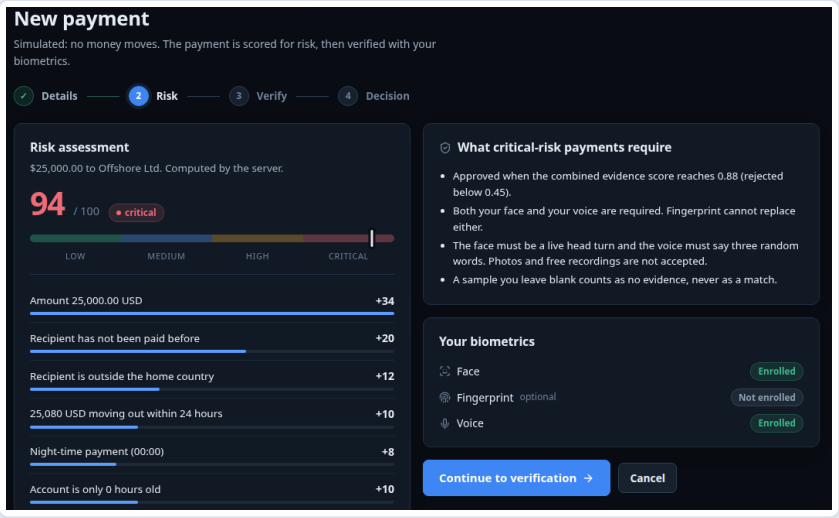
Risk assessment

The server scores the payment, lists every factor, and says what this level requires.

Decision with evidence

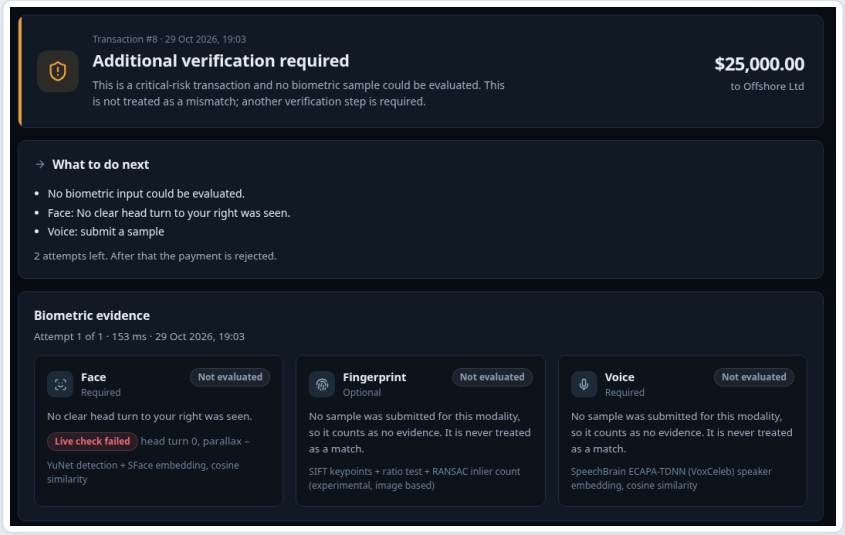
Match confidence, input quality and a plain reason. A blank sample is no evidence, never a match.

A large payment asks for more



Critical risk

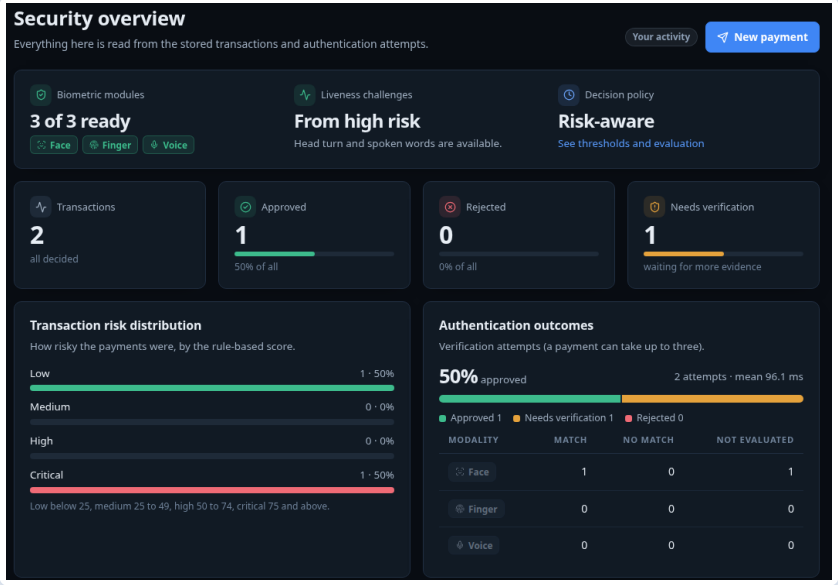
Face and voice are both required, and both must be live. Fingerprint cannot replace either.

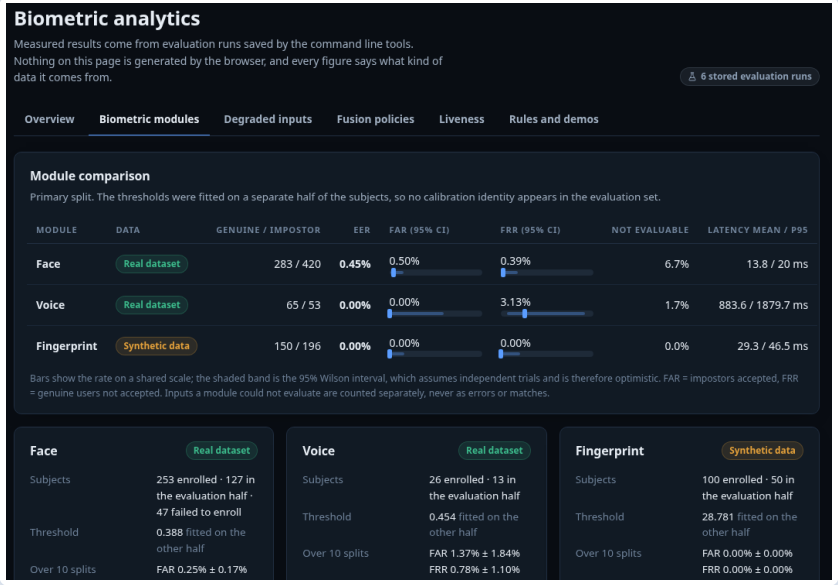


A still camera is refused

No head turn was seen, so it asks for more evidence and lists exactly what to do. Three attempts.

Seeing what happened





Dashboard and history

Totals, risk and outcomes from stored data, plus history.

Analytics

Every figure is labelled real, synthetic or simulated.

How it is built and tested

Frontend	React, Vite, Recharts
Backend	FastAPI, SQLAlchemy, SQLite
Face	OpenCV: YuNet + SFace
Voice	SpeechBrain ECAPA-TDNN
Spoken words	faster-whisper (tiny.en)
Fingerprint	SIFT keypoints + RANSAC

323 automated tests passing

1 browser test of the whole flow in real Chrome

10 random splits per evaluation

Models are open source and run on a normal CPU. No paid services.

How we evaluated it

Public datasets
LFW faces, LibriSpeech voices

→

Trials
genuine and impostor

→

Split people
two halves, never mixed

→

Fit on half A
thresholds

→

Measure on half B
FAR, FRR, EER

→

Repeat
10 random splits

Real data

Faces from LFW and voices from Mini LibriSpeech, public datasets of real people.

Synthetic data

Fingerprints are generated. We found no real fingerprint dataset we could download.

What we report

FAR, FRR and EER with 95% confidence intervals, plus how many inputs could not be evaluated.

Results: each biometric on its own

	Data	Trials	EER	FAR (95% CI)	FRR (95% CI)	Time
Face	Real (LFW)	283 / 420	**0.45%**	0.50% (0.1 to 1.8)	0.39% (0.1 to 2.2)	14 ms
Voice	Real (LibriSpeech)	65 / 53	**0.00%**	0.00% (0.0 to 6.9)	3.13% (0.9 to 10.7)	884 ms
Fingerprint	Synthetic	150 / 196	**0.00%**	0.00% (0.0 to 1.9)	0.00% (0.0 to 2.5)	29 ms

Trials are genuine / impostor attempts. EER: false accepts equal false rejects. FAR: impostors accepted. FRR: real users rejected. CI: 95% confidence interval, in %.

0.45%

face equal error rate

127 people in the evaluation half

1.4% ± 1.8%

voice false accepts over 10 splits

only 13 speakers per half

When input is poor, and the liveness checks

It refuses to guess

Face: mild blur	**35% refused**
Face: dim light	**68% refused**
Face: strong blur, dark, low resolution	**100% refused**
Voice: noise at 10 dB	**14% real users rejected**
Voice: noise at 0 dB	**98% refused**

Too-poor input is refused instead of scored. The user is asked to retry.

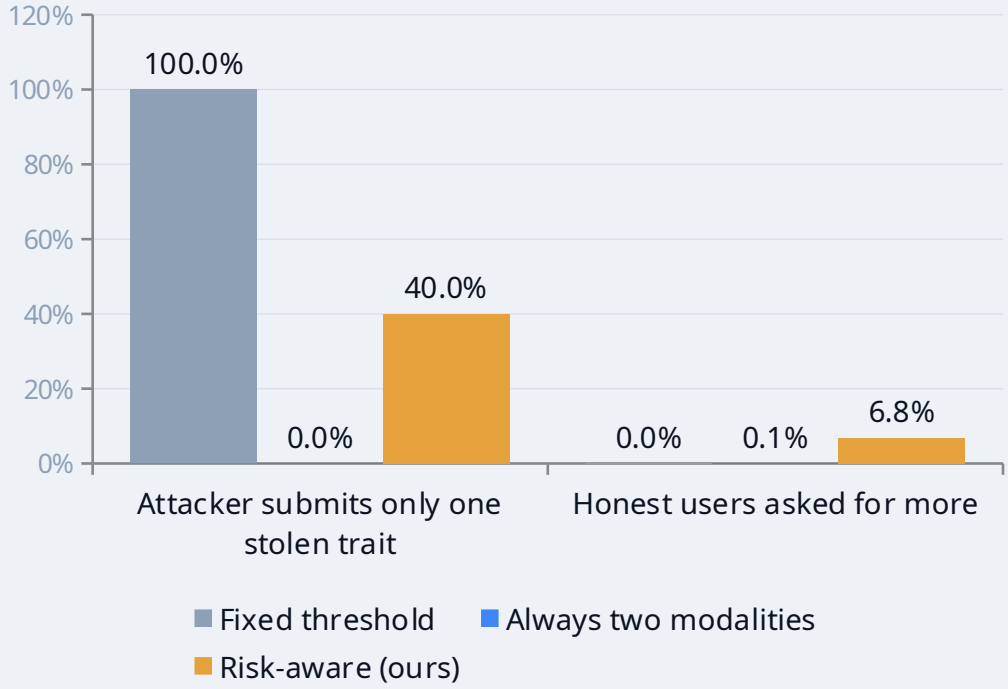
Liveness checks

simulated attacks

Tilted flat photo passes	**1 in 1000**
Two-photo swap passes	**0 in 192**
Spoken words heard (clean audio)	**95%**
Spoken words heard (10 dB noise)	**91%**
Unrelated recording passes	**0 in 650**

Face attacks are simulated from photos. The spoken-word test uses real speech.

Results: fixed threshold vs risk-aware



Simulated attempts from measured scores. Fingerprint scores are synthetic.

Plain threshold

Fooled every time when only one trait is submitted.

Always two

Safe, but a good user with one trait is blocked.

Risk-aware

Closes the hole for medium risk and up. Low-risk payments still accept one trait, so 40% get through.

Limitations and what comes next

What we have not proven

- Fingerprint was only tested on generated prints
- Datasets are small and easy, so real-world error will be higher
- Liveness does not stop videos, deepfakes or voice cloning, and was never run on a real person turning their head
- Prototype: simulated money, no HTTPS, SQLite

What comes next

- Real fingerprint sensor and dataset
- Public spoofing datasets to test liveness properly
- On-device biometrics (WebAuthn) and HTTPS
- Larger multi-session datasets and tuned thresholds

Conclusion

1. **The check should match the risk.**
   TriLock raises the bar for large payments and keeps small ones smooth.

2. **Missing evidence is never a match.**
   Failed or missing input asks for more, it does not approve.

3. **We measured it, and said where it is weak.**
   Real and synthetic data are labelled, and the limits are listed.



Thank you

Questions?